Tilemachos Bolioudakis

PhD Candidate

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Education

2021–present **PhD**, *Cornell University*, Civil & Environmental Engineering, Environmental C.F.D.

2021-2024 **M.Sc. in Environmental Fluid Mechanics and Hydrology**, *Cornell University*

2019-2021 **M.Sc. in Oceanography and Management of the Marine Environment**, *NKUA*

2012-2019 **B.Sc. in Physics**, *National and Kapodistrian University of Athens (NKUA)*, Section of Environmental Physics and Meteorology

Experience

08/2024 - present **System Administrator**, *Environmental C.F.D. Group*, Civil & Environmental Engineering, Cornell University

01/2023 - present **Graduate Teaching Assistant**, *Civil & Environmental Engineering*, Cornell University, Teaching CV attached

08-12/2022 **Graduate Research Assistant**, *Civil & Environmental Engineering*, Cornell University

08/2021 - 08/2022 **Cornell Fellow Graduate Research Assistant**, *Civil & Environmental Engineering*, Cornell University

07-11/2020 **Volunteer**, *Youth Directorate*, Ministry of Education / Religious Affairs, Hellenic Republic

Awards

08/2022- present **Conference Travel Grants**, *ISSF 2022*, AGU-OSM 2024, AGU-OSM 2026
Cornell Graduate School

05-08/2023 **DeFrees Fellowship**, *Civil & Environmental Engineering*, Cornell University

2022 **Gerondelis Foundation Scholarship**, *Graduate Study Scholarship Program*, Gerondelis Foundation Inc., Lynn, MA, United States

2021 **Cornell University Fellowship**, *Academic Year 2021-2022*

2020 **Scholarship for M.Sc. Tuition Fees**, *Based on academic achievements*, National and Kapodistrian University of Athens

Publications

03/2026 **Published**, *Journal of Physical Oceanography*, American Meteorological Society, DOI: https://doi.org/10.1175/JPO-D-24-0181.1

10/2025 **Preprint by Research Square**, *https://doi.org/10.21203/rs.3.rs-7906656/v1*

Invited Talks / Conference Presentations

03/2026 Presentation **16th Graduate Research Symposium**, *Civil & Environmental Engineering*, Cornell University

02/2026 Poster **American Geophysical Union**, *Ocean Sciences Meeting*, Glasgow, Scotland, UK, PS24C-1754

07/2024 Invited Talk **Interinstitutional Postgraduate Program NKUA-HCMR**, *Pathways to turbulence formation in high amplitude internal solitary waves of depression shoaling over gentle slopes*, Dep. of Physics, Athens, Greece, Announcement link

05/2024 Poster **Fluid Dynamics of Sustainability and the Environment**, *Summer School*, Ecole Polytechnique, Paris, France, http://fdse.org

02/2024 Poster **American Geophysical Union**, *Ocean Sciences Meeting*, New Orleans, Louisiana, USA, https://agu.confex.com/agu/OSM24/meetingapp.cgi/Person/1469507

11/2023 Presentation **American Physical Society**, *76th Annual Meeting of the Division of Fluid Dynamics*, Washington, D.C., USA, https://meetings.aps.org/Meeting/DFD23/Session/T44

11/2022 Presentation **American Physical Society**, *75th Annual Meeting of the Division of Fluid Dynamics*, Indianapolis, Indiana, USA, https://meetings.aps.org/Meeting/DFD22/Session/L27.2

08/2022 Poster **9th International Symposium on Stratified Flows**, *Cambridge,UK*, https://www.damtp.cam.ac.uk/events/issf2022/

Approved High-Performance-Computing Proposals

09/2024 **Maximize Project**, *NSF supported ACCESS Allocations, USA*, Role: Co-PI, Amount: 6.5M Service Units

03/2021 **Preparation/Development Projects**, *GRNET HPC System A.R.I.S., Greece*, Role: Co-PI, Amount: 200K Service Units

Additional Training

05/2024 **Fluid Dynamics of Sustainability and the Environment**, *Summer School*, Ecole Polytechnique, Paris, France, http://fdse.org

12/2020 **GRNET PRACE Training Center**, *Efficient Use of HPC Systems*

11/2020 **GRNET PRACE Training Center**, *Introduction to Parallel Programming*

10/2018 **Hellenic Physics Union Summer School**, *Aegina,Greece*

07/2018 **25th summer school-conference. Dynamical systems and complexity**, *NCSR Demokritos*, Athens, Greece

Languages

English British Council IELTS (grade 7.0 of 9) *05/2019*

Greek Native Language

Computer Skills

Experience Linux/Unix environments, High-Performance Computing

Data Synology NAS servers, Globus data transfers

Software MATLAB, TecPlot, VisIt, LaTeX, Microsoft Office, ArcGIS

Programming Fortran, Python, R

Teaching CV

CEE (ENGRD) 3200 Engineering Computation (Undergraduate)

Teaching Assistant: Spring 2023, 2024, 2025, 2026 (ongoing)

- Led weekly recitations and review sessions supporting homework and project development.
- Held regular and exam-focused office hours for group and one-on-one guidance.
- Designed and graded problem sets, projects, and exams integrating course material with engineering applications.
- Developed computational examples connecting theory with numerical implementation.
- Responded to student questions and provided clarifications through Piazza.

CEE 3310 Fluid Mechanics (Undergraduate)

Teaching Assistant: Fall 2023, 2024

- Supported lecture discussions through board-based problem solving.
- Assisted in exam preparation and laboratory experiment design.
- Led laboratory sessions in the DeFrees Teaching Laboratory flumes and supervised junior TAs.
- Graded assignments and exams with detailed written feedback.
- Provided online clarifications through Canvas.

CEE 6000 Numerical Methods for Engineers (Graduate)

Teaching Assistant: Fall 2025

- Led advanced office hours on numerical modeling and stability analysis.
- Assisted in assignment design and grading.

Private Tutor (High School Level), Athens, Greece

During B.Sc. and M.Sc. studies

- Tutored students in Physics, Mathematics, and Chemistry, including preparation for final and university entrance examinations.